

Infinity Hotel Check-In Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	infinity-hotel-check-in

Big idea: Mathematical concepts can be coherent even when they violate ordinary experience.

Student-Facing Prompt

Inspect Infinity Hotel Check-In Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Infinity Hotel Check-In Stimulus A	Student-facing stimulus 1: One new guest arrives at a full infinite hotel; every current guest moves by $n \rightarrow n + 1$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Infinity Hotel Check-In Stimulus B	Student-facing stimulus 2: Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Infinity Hotel Check-In Reveal / Twist	Student-facing stimulus 3: Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical concepts can be coherent even when they violate ordinary experience.

Actual Lesson Questions

#	Question for students
1	After inspecting Infinity Hotel Check-In Stimulus A, what is your first knowledge claim?
2	Which exact detail from Infinity Hotel Check-In Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Can mathematics know beyond intuition?

Ready-to-Use Examples

- One new guest arrives at a full infinite hotel; every current guest moves by $n \rightarrow n + 1$.
- Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Assign students numbered hotel rooms and announce that every room is occupied.
2. Introduce one new guest and ask whether the hotel can make room.
3. Use the rule $n \rightarrow n + 1$ so each guest moves to the next room.
4. Discuss why the model works for countably infinite rooms but not for a finite hotel.
5. Debrief the difference between everyday imagination and formal definition.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Can mathematics produce knowledge about things we cannot imagine concretely?
- Why can a coherent formal system feel impossible?
- What is the role of intuition in mathematical knowledge?

Knowledge Questions

- Can mathematics know beyond intuition?
- What makes a mathematical object meaningful?
- When should intuition be trusted in mathematics?

Transfer Example

For example, run this activity with a simple classroom version using numbered room cards.. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...